

KREPO

Bridging the Institutional Knowledge Gap through Automated Context-Aware Onboarding

Note: This is an AI generated report but the salient points were provided in detail in prompt & reviewed later.

Abstract

In the modern high-growth engineering environment, information propagation remains one of the primary bottlenecks to organizational velocity. As teams scale beyond their early staff sizes, technical debt often transitions from code-based issues to "knowledge debt"—the fragmentation of critical system context across disparate tools and individuals. KREPO is an AI-native solution designed to ingest multi-source organizational data (GitHub, Slack, Jira, Drive, and Meeting Recordings) to create dynamic, context-aware onboarding modules. By utilizing a Retrieval-Augmented Generation (RAG) architecture and LLM agents, KREPO automates the technical onboarding process, significantly reducing the instructional burden on senior engineers and ensuring that institutional knowledge is preserved and accessible.

1. The Problem Space

1.1 Knowledge Breakdown in Scaling Teams

As organizations grow, the "surface area" of information increases exponentially. Traditional methods of information propagation—such as word-of-mouth or static documentation—fail to scale. In teams of 10+, knowledge silos begin to form, leading to duplicated efforts and "stale" information that hinders decision-making.

1.2 The Senior Engineer Bottleneck

Senior engineers frequently face "technical onboarding fatigue." They are often required to repeat the same high-level system architectural explanations to every

new hire, diverting their focus from critical shipping targets and high-impact architectural work. This repetitive cycle is not only inefficient but also represents a significant hidden cost in engineering hours.

1.3 Contextual Disconnect

Standard training programs often lack the "why" behind technical decisions. They convey the "how" (e.g., how to run the build script) but fail to convey the broader company context or the historical evolution of the codebase. Consequently, new hires may understand the syntax of the system without understanding its soul.

1.4 Non-Technical Overestimation

A recurring friction point exists where non-technical stakeholders (PMs, Sales, Leadership) overestimate the transparency of current systems. They often assume that if data is in "the cloud," it is easily discoverable. This leads to a persistent querying of senior engineers for high-level system design questions that should be available via self-service.

2. Current Alternatives (State of the Market: 2026)

Current solutions are largely fragmented or reliant on human-heavy processes:

- **Outsourced Module Onboarding:** Larger firms often hire external agencies to create training modules. However, these quickly become obsolete as the codebase evolves weekly.
- **Synchronous Training Sessions:** Teams typically rely on 6–7 in-person or virtual meetings for technical onboarding, creating a heavy scheduling burden and information retention issues.
- **Manual Querying:** Non-technical staff are forced to interrupt engineering workflows to understand system capabilities, leading to context-switching costs across the entire technical department.

3. Our Solution: KREPO

KREPO acts as the "Institutional Brain," synthesizing raw data into structured learning pathways.

3.1 Multi-Source Information Ingestion

KREPO does not rely on a single source of truth. It integrates directly with the tools where work actually happens:

- **Version Control:** GitHub repositories for code structure and commit history.
- **Written Documentation:** Confluence and Notion pages.
- **Project Management & Communication:** Slack threads and Jira tickets to capture decision-making "breadcrumbs."
- **Cloud Storage:** Deep integration with Google Drive.
- **Verbal Communication:** Transcription of meeting recordings to capture nuanced verbal context.

3.2 Security and Governance

Understanding the sensitivity of corporate data, KREPO is built with a focus on:

- **SOC2 Compliance:** Standard-grade security protocols for data handling.
- **Role-Based Access Control (RBAC):** Ensuring users can only query information they have explicit permission to access within the source systems.

3.3 Dynamic Module Creation

Unlike static wikis, KREPO generates user-specific and task-specific modules. If a developer is assigned a specific Jira ticket, KREPO can generate a localized onboarding module containing only the relevant code paths, past Slack discussions, and documentation needed to complete that specific task.

4. Implementation & Technical Architecture

The KREPO engine utilizes a modern AI stack designed for low latency and high relevance.

Layer	Technology
Orchestration	LangChain (RAG)
Backend API	Python (FastAPI)
Frontend	React.js
Vector Database	QDrant
Speech-to-Text	Lemonfox (Audio Transcription)
Reasoning Engine	Claude 3.5 Sonnet (Agentic workflows)
Integrations	Custom OAuth connectors for Slack, Jira, GitHub

4.1 The RAG Pipeline

The core of KREPO is a sophisticated Matryoshka Retrieval-Augmented Generation (RAG) pipeline. Raw data is chunked and embedded into QDrant. When a query is made, the Claude Sonnet agent doesn't just look for keywords; it analyzes the *intent* and retrieves the most relevant context across all integrated platforms to formulate a response or build a training module.

5. Market Analysis & Differentiation

The enterprise knowledge space is evolving rapidly. A notable milestone was Workday's acquisition of **Sana.ai** in September 2025. While Sana pioneered AI-driven learning, KREPO differentiates itself by:

1. **Technical Depth:** Focusing specifically on the "Engineering Knowledge Gap" rather than general HR onboarding.
2. **Task-Specific Logic:** Generating modules based on active tickets/tasks rather than just high-level company overviews.
3. **Cross-Tool Synthesis:** Moving beyond simple document search to synthesize "unstructured" knowledge from Slack and meeting transcripts.

6. Results & Validation

- **MVP Development:** Successfully built a functional prototype capable of ingesting GitHub and Slack data to answer architectural questions.
- **Market Feedback:** Demonstrated the MVP to a selection of personal industry contacts (Senior Leads and PMs). The feedback was overwhelmingly positive, specifically regarding the reduction in "onboarding meeting" requirements.

7. Future Roadmap

The next phase of KREPO development focuses on optimizing core systems and broadening accessibility. This includes refining the ingestion and querying mechanisms for higher precision, transitioning the platform from a standalone web-app to an integrated enterprise application, and implementing scalability adjustments. Additionally, the project will seek formal clearance to transition from a prototype into a full-scale entrepreneurial endeavor.

Find visual demo in /documentation directory.